FINAL YEAR PROJECT
SYNOPSIS



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1. INTRODUCTION

Personal belongings are lost every day in colleges, offices, and public places. Recovery today relies almost entirely on notice boards, social media groups, and manual word-of-mouth. Existing digital lost-and-found tools mostly provide a searchable list of lost and found reports, but finding the right match among many similar-looking entries remains genuinely difficult, and confirming that a claimant is the true owner of a recovered item is even harder.

LostLink is proposed as an intelligent lost-and-found system that combines image, text, location, and time information to surface likely matches automatically, and adds a separate, evidence-based verification step to confirm that a claimant is the real owner rather than simply the first or most convincing person to respond. The system is intended to reduce the time and effort required to recover lost belongings while making the recovery process transparent and auditable for administrators.

This synopsis presents the problem being addressed, the systems that already exist in this space, the specific gaps LostLink is designed to fill, and the proposed methodology, architecture, and evaluation plan for the project.

2. PROBLEM STATEMENT

Several distinct difficulties make lost-and-found recovery unreliable in practice:

- Relevant lost or found reports are difficult to find among many listings.
- Keyword search misses items that are visually similar but described in different words.
- Two people frequently describe the same object very differently.
- Location and time context is collected by existing systems but rarely used in matching.
- Multiple people may claim the same found item, with no reliable way to compare their claims.
- Manually verifying legitimate ownership is slow, inconsistent, and depends entirely on administrator judgement.

How can an intelligent system identify potential matches and verify legitimate ownership, while reducing false and fraudulent claims?

3. MOTIVATION

The motivation behind the project is as follows:

- Reduce the time required to recover lost belongings.
- Reduce dependence on manual searching across notice boards and social media groups.
- Improve matching accuracy by using multiple information sources together — image, text, location, and time — instead of keywords alone.
- Reduce fraudulent or incorrect ownership claims through a structured verification mechanism.
- Provide one centralized recovery workflow instead of scattered, informal channels.
- Keep the recovery process transparent and auditable, so administrators can review why a match or a claim was accepted.

4. EXISTING SYSTEM

A review of current lost-and-found and image-matching literature shows that several relevant systems already exist, though each addresses only part of the problem LostLink is designed to solve:

System	What It Does	Where It Falls Short
Dhanawardhana et al. (2025)	MobileNetV2-based Siamese network, fine-tuned on 848 bag-image pairs; fuses image, text, location and time similarity into one score.	Fine-tuning ties the model to a single item category (bags); no ownership-verification component.
LostNet (Zhou et al., 2024)	MobileNetV2 with CBAM attention, fine-tuned for image matching, combined with perceptual hashing.	Image-only; single-category focus; no text, location, time, or verification signal.
Campus claim-arbitration system (2026, preprint)	Describes an ownership-verification and multi-claim arbitration workflow for campus lost-and-found.	Non-peer-reviewed; no quantified fraud-detection or verification-accuracy metrics published.
Typical campus lost-and-found portals	Registration, keyword/category search, notifications, and an administrator panel.	Matching is keyword or category based; the correct item is easy to miss among similar listings.

5. RESEARCH / PROJECT GAP

Combining image, text, location and time for lost-and-found matching is not, by itself, a new idea — it has already been demonstrated by Dhanawardhana et al. (2025). LostLink does not claim to invent multi-factor matching; its contribution is narrower, more specific, and testable, and consists of the two gaps described below.

5.1 Gap 1 — What Kind of Pretrained Model Actually Matches Items

Existing matching systems (Dhanawardhana et al., LostNet) are built on classification-pretrained convolutional networks such as MobileNetV2 and ResNet50. These networks were originally trained to sort images into a fixed set of categories, not to compare two arbitrary photos for similarity. To repurpose them for matching, prior work had to fine-tune a Siamese network on hand-labelled pairs of same-item and different-item photographs — for example, 848 bag images in Dhanawardhana et al.'s study. This fine-tuning step ties the resulting model to whichever category it was trained on; extending it to a new item category would require collecting a new labelled dataset and retraining from scratch.

LostLink instead uses similarity-native embedding models — CLIP and DINOv2 — which are pretrained so that visual similarity is the direct objective, not a side effect of classification. These models can be used zero-shot, without any category-specific fine-tuning, and are designed to generalise across a broad range of object categories. LostLink evaluates whether this removes the category-specific fine-tuning bottleneck seen in prior work, and measures matching accuracy across multiple item categories rather than one.

5.2 Gap 2 — Is Ownership Verification Actually Measured Anywhere

A visual or textual match is not proof of ownership: it is common for more than one person to claim the same found item, for example a black backpack. Existing work either does not address this problem at all, or describes a claim-arbitration workflow conceptually (as in the 2026 campus preprint) without reporting any measured accuracy for it. No peer-reviewed source identified in this

review publishes quantitative fraud-detection results for a lost-and-found ownership-verification mechanism.

LostLink addresses this by building a scored, evidence-based verification step, in which a claimant is asked for private identifying details that were not publicly displayed with the listing — such as a distinguishing mark, the contents of a bag, or purchase information. Each claim is assigned a verification score, and the mechanism is evaluated the way a classifier would be: using False Acceptance Rate (how often a fraudulent claimant is wrongly approved) and False Rejection Rate (how often a genuine owner is wrongly denied), measured at varying score thresholds. Where several people claim the same item, the same scoring mechanism is used to rank competing claims, and the ranking accuracy is evaluated in the same way.

6. PROPOSED SYSTEM

LostLink is proposed as a web-based platform in which users report lost and found items with a photograph, category, description, location, and approximate time. The system automatically compares new reports against existing listings using a combination of image, text, location, and time similarity, and presents ranked candidate matches with a transparent, explainable score breakdown. Once a likely match is identified, a claimant must pass an evidence-based ownership-verification step before an item is released, with administrators able to review borderline or disputed cases.

The system is organised into the following modules:

- User registration and authentication
- Lost-item reporting
- Found-item reporting
- Image embedding generation (CLIP / DINOv2)
- Text similarity analysis
- Location and time matching
- Multi-factor match-score generation
- Potential-match notification
- Ownership verification
- Claim management and ranking
- Administrator verification dashboard
- Secure handover and recovery history

7. OBJECTIVES

7.1 Primary Objectives

- Develop a centralised lost-and-found platform.
- Automatically surface potential matches from new reports.
- Use pretrained computer-vision models for image similarity.
- Combine visual and contextual signals — text, location, and time.
- Build a scoring-based ownership-verification mechanism.
- Reduce false and fraudulent claims.

7.2 Research Objectives

RQ1 — How do zero-shot embeddings (CLIP, DINOv2) compare to trained, category-specific fusion models for cross-category lost-and-found matching?

RQ2 — Can a scored, evidence-based verification mechanism measurably lower the false-acceptance rate for fraudulent claims without a large rise in false rejections of genuine owners?

8. SCOPE OF THE PROJECT

8.1 In Scope

- Web application front end (React).
- Lost and found reporting with image upload.
- Image similarity using pretrained CLIP / DINOv2 embeddings.
- Text, location, and time similarity.
- Multi-factor match-score generation.
- Ownership verification and scoring.
- Multiple-claim ranking.
- Notifications for potential matches and claim status.
- Administrator dashboard for review and approval.
- Recovery history and audit trail.

8.2 Future Scope

- Mobile application.
- QR-code-based item registration.
- Integration with campus security systems.
- A cross-campus recovery network.
- Fine-tuning of embedding models if a sufficiently large dataset becomes available.

9. METHODOLOGY

The proposed matching and verification pipeline follows five steps:

1. Report

The user submits an image, category, description, location, and approximate date/time for a lost or found item.

2. Embed

The item photograph is passed through a pretrained image model (CLIP or DINOv2) to produce a numerical embedding vector.

3. Match

Cosine similarity between image embeddings is combined with text, location, and time similarity into a single weighted score.

4. Verify

A claimant is asked for private identifying evidence not shown in the public listing; each answer contributes to a verification score.

5. Decide

High-confidence matches and verifications are recommended automatically; medium-confidence cases are sent for administrator review; low-confidence cases are rejected.

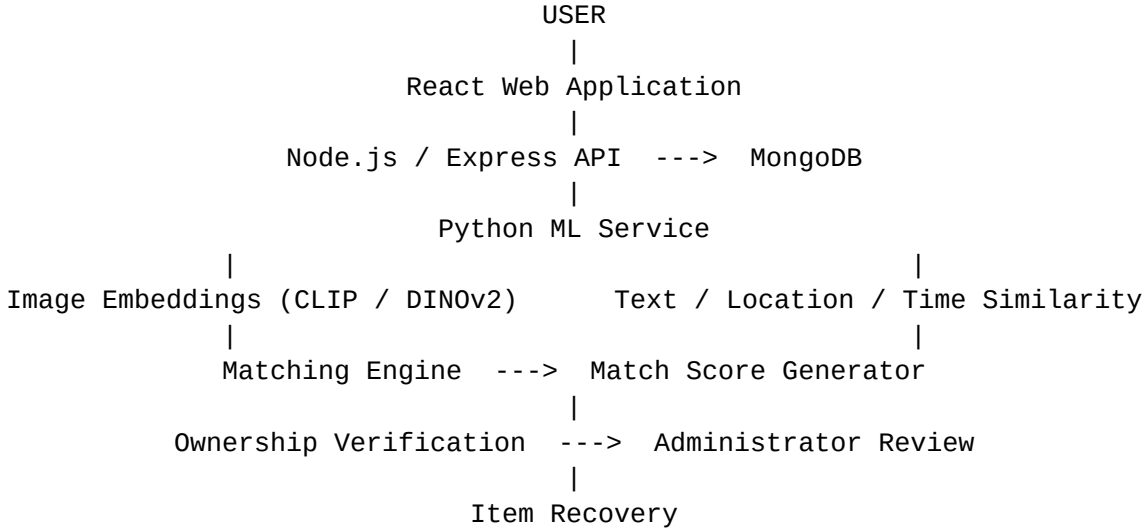
$$\textbf{Final Score} = w1 \cdot \textbf{Image} + w2 \cdot \textbf{Text} + w3 \cdot \textbf{Location} + w4 \cdot \textbf{Time}$$

9.1 Evaluation

Matching performance is evaluated using precision, recall, F1-score, and top-K retrieval accuracy, comparing CLIP, DINOv2, and a keyword/category baseline. Ownership-verification performance is evaluated using False Acceptance Rate and False Rejection Rate at varying score thresholds, and multiple-claim ranking is evaluated by the proportion of simulated cases in which the genuine owner is ranked first.

10. SYSTEM ARCHITECTURE / BLOCK DIAGRAM

The system follows a layered architecture in which the client application communicates with a Node.js API, which in turn delegates image and text similarity computation to a dedicated Python machine-learning service:



11. TECHNOLOGIES / TOOLS USED

Category	Tools / Technologies
Frontend	React.js, Tailwind CSS
Backend	Node.js, Express.js
Database	MongoDB
ML Service	Python
Image Processing	OpenCV
Image Embeddings	CLIP, DINOv2
Similarity Computation	Cosine similarity

Category	Tools / Technologies
Text Matching	TF-IDF / sentence embeddings
Maps	Leaflet + OpenStreetMap, or Google Maps API
Authentication	JWT
Notifications	Firebase Cloud Messaging, Email
Version Control / Deployment	Git, GitHub; Vercel (frontend); Render/Railway (backend); MongoDB Atlas

12. PROJECT TIMELINE / GANTT CHART

Phase	Aug	Sep	Oct	Nov	Dec	Jan
Literature Review	X	X				
Requirement Analysis	X	X				
UI / UX Design		X	X			
Backend & Database			X	X		
ML / Matching Module			X	X	X	
Ownership Verification				X	X	
Frontend Integration				X	X	
Testing					X	X
Evaluation					X	X
Research Paper Draft					X	X
Final Documentation						X

13. REFERENCES

[1] B. M. P. Dhanawardhana et al., "Enhancing Lost and Found Systems with Multi-Modal Deep Learning: Integrating SBERT and Siamese Networks for Improved Semantic Matching," Advances in Artificial Intelligence and Machine Learning, vol. 5, no. 2, pp. 3736–3754, 2025.
Key point: Fuses image, text, location and time using a Siamese network fine-tuned on 848 bag images; closest prior work to LostLink, but limited to one item category.

[2] M. Zhou et al., "LostNet: A Smart Way for Lost and Find," PLOS ONE, 2024 (arXiv:2301.02277).
Key point: Image-only matching using MobileNetV2 and perceptual hashing; reports high accuracy but does not use text, location, or time.

[3] A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," in Proceedings of ICML, 2021.
Key point: Introduces CLIP, the zero-shot image-text embedding model used for image similarity in LostLink.

[4] M. Oquab et al., "DINOv2: Learning Robust Visual Features without Supervision," arXiv:2304.07193, 2023.

Key point: Introduces DINOv2, a self-supervised image-embedding model used as the second image-similarity option in LostLink.

[5] T. Khandelwal and S. Poria, "Multimodal Embeddings for Image-Text Retrieval: A Survey," ACM Computing Surveys, 2022.
Key point: Surveys multimodal embedding techniques; used as background for combining image and text similarity.

[6] "Smart Lost and Found System for Campus," Zenodo preprint, 2026.
Key point: Describes an ownership-verification and claim-arbitration workflow for campus lost-and-found, but reports no measured accuracy for it.

[7] "Uni Find: Lost and Found Objects Management System for Campus using AI Cloud Services," ResearchGate, 2025.
Key point: Uses cloud AI services and ID-based login for campus lost-and-found; verification is identity-based, not evidence-based.

[8] "An Effective Lost and Found System in University Campus," International Journal of Advanced Research, vol. 8, no. 32, pp. 99–112, 2023.
Key point: Survey-based study showing most lost items on campus are never recovered; used to support the problem statement.

[9] "AI-Driven Lost and Found Management System for Institutions," ResearchGate, 2026.
Key point: General AI-based institutional lost-and-found system; cites CLIP but does not evaluate it against alternatives.

[10] G. Koch, R. Zemel, and R. Salakhutdinov, "Siamese Neural Networks for One-shot Image Recognition," ICML Deep Learning Workshop, 2015.
Key point: Foundational paper on Siamese networks, the architecture used by prior lost-and-found matching systems.

[11] "Lost Item Identification Model Development Using Similarity Prediction Method with CNN ResNet Algorithm," ResearchGate, 2023.
Key point: Reports only ~30% accuracy using ResNet50 with Euclidean distance, motivating the use of similarity-native embeddings instead.

[12] "Progressive Multi-Subspace Fusion for Text-Image Matching," Complex & Intelligent Systems, 2025.
Key point: General technique for combining image and text similarity; used as background for the match-score fusion design.